

Brain data

Brain structure data were collected from the literature [1-7], and include overall brain and body size, prefrontal cortex gray and white matter for left and right hemisphere, frontal motor cortex gray and white matter for left and right hemisphere, posterior neocortex (overall neocortex minus frontal lobe), cerebrocerebellum (cerebellar hemispheres), spinocerebellum (vermis + paravermis), striatum, hippocampus, entorhinal cortex, diencephalon, mesencephalon, olfactory bulb, paleocortex (includes retrobulbar cortex, prepiriform cortex, lateral medial and internal olfactory tracts, anterior commissure, olfactory tubercle, substantia innominata, and basal nucleus of Meynert), septum, and medulla. Prefrontal and frontal motor cortex were delineated using a bootstrap approach based on the cytoarchitectonic delineation of the frontal lobe (defined by cell shape, cell density, cell size, presence of granular layer IV, diffuseness of white/gray matter border, and lamination patterning) [2-4]. This bootstrap approach consists of considering cumulative volumes along the anterior axis of the frontal cortex. For the current study, volumes of the 7th cumulative interval along the anterior axis of the frontal cortex are considered following evidence of a marked grade shift in the white to gray matter ratio between Old World monkeys, great apes, and humans [2, 4]. This procedure may underestimate prefrontal cortex volume in humans considering evidence of a stabilization of the absolute volume of the primary motor cortex (BA4) in humans compared to great apes [8], although no information is currently available on the size of the premotor cortex. Brains used in the current analysis are housed at the C.&O. Vogt Institute for Brain Research (University of Düsseldorf, Germany) [5]. The following individuals (N=34) were included in the analyses (* indicates females): *Homo sapiens* (collection numbers pm2, pm4, pm6, pm7, pm13, pm1*, pm8*, pm9*, pm10*), *Pan troglodytes* (280, 497*), *Gorilla gorilla* (375, 8214*), *Hylobates lar* (1203, 397*, 81146*), *Cercopithecus mitis* (261), *Cercopithecus ascanius* (219*), *Erythrocebus patas* (1545, 1341*), *Miopithecus talapoin* (1171, 1201*), *Lophocebus albigena* (242, 221, 99*), *Papio anubis* (97*), *Piliocolobus badius* (213*), *Nasalis larvatus* (1365*), *Alouatta seniculus* (1184), *Ateles geoffroyi* (1000), *Lagothrix lagotricha* (1571*), *Cebus albifrons* (1200, 6062*), and *Pithecia monachus* (1180).

Phylogenetic mapping

To infer the evolutionary pathways on each branch of the tree of life covered by our sample we employ a novel comparative framework [9] using an approach that is based on the principles of an adaptive peak model of evolution [10, 11]. This framework comprises a comparison of lineage-specific rates of evolution revealing all possible combinations of trait increase/decrease between two traits. ESM 2 provides detailed information on the results of these analyses. In each Figure of ESM 2 the lineage specific rates of evolution of a mosaic pattern are compared to those of overall brain size. The lineage-specific rates of evolution are estimated using the method described above [10]. The mosaic patterns are identified by a phylogenetically controlled principal components analysis [12, 13]. For each mosaic pattern each figure displays two types of results (the rates of evolution for each branch and the values for each node in the tree) and two types of visualizations (indicating a joint increase in brain size and mosaic pattern on the left hand side and indicating variation in mosaic patterns for brain size increase versus decrease on the right hand side). The two visualizations relate to the same data but highlight different aspects of the results. The visualization on the left indicates a lack of increase in both the mosaic pattern and brain size, or an increase in brain size but not in mosaic pattern, as white, an increase in the mosaic pattern but not brain size as yellow, and an increase in both the mosaic pattern and brain size as red. The visualization on the right indicates variation in the mosaic pattern (either increase or decrease) as red when brain size decreases and as green when brain size increases.

A comparison of the evolutionary history of each size-independent mosaic pattern with absolute brain size is preferred over a comparison with relative brain size because the latter is confounded by body size specific selection pressures [9]. A decrease in body size that is not paralleled by a proportionate decrease in brain size leads to an increase in relative brain size; a trend that reflects selection on body size rather than brain size. PC1 scores confirm this to be predominantly present in our sample with lineages that have undergone phyletic dwarfing in body size indicating high scores for relative brain size. Moreover, our aim is to quantify the dynamic between the mosaic increase of a particular organizational component and the overall size of the mosaic component. For these purposes, only a comparison with absolute brain size is suitable.

1. Smaers J.B., Steele J., Zilles K. 2011 Modeling the evolution of cortico-cerebellar systems in primates. In *New Perspectives on Neurobehavioral Evolution* (eds. Johnson J.I., Zeigler H.P., Hof P.R.), pp. 176-190. New York, Annals of the New York Academy of Sciences.
2. Smaers J.B., Steele J., Case C.R., Cowper A., Amunts K., Zilles K. 2011 Primate prefrontal cortex evolution: human brains are the extreme of a lateralized ape trend. *Brain, Behavior and Evolution* **77**, 67-78.
3. Smaers J.B., Schleicher A., Zilles K., Vinicius L. 2010 Frontal white matter volume is associated with brain enlargement and higher structural connectivity in haplorrhine primates. *PLoS One* **5**(2), e9123.
4. Smaers J.B., Mulvaney P.I., Soligo C., Zilles K., Amunts K. 2012 Sexual dimorphism and laterality in the evolution of the primate prefrontal cortex. *Brain, Behavior and Evolution* **79**(3), 205-212.
5. Zilles K., Amunts K., Smaers J.B. 2011 Three brain collections for comparative neuroanatomy and neuroimaging. In *Resources and Technological Advances for Studies of Neurobehavioral Evolution* (eds. Johnson J.I., Zeigler H.P., Hof P.R.), pp. E94-E104. New York, Annals of the New York Academy of Sciences.
6. Smaers J.B., Steele J., Case C.R., Amunts K. In Press Laterality and the evolution of the prefronto-cerebellar system in anthropoids. In *The evolution of human handedness* (eds. McGrew W.C., Marchant L., Schiefenhövel W.). New York, Annals of the New York Academy of Sciences.
7. Stephan H., Frahm H., Baron G. 1981 New and revised data on volumes of brain structures in insectivores and primates. *Folia Primatologica* **35**(1), 1-29.
8. Frahm H.D., Stephan H., Stephan M. 1982 Comparison of brain structure volumes in Insectivora and Primates. 1. Neocortex. *Journal für Hirnforschung* **23**(4), 375-389.
9. Smaers J.B., Dechmann D.K.N., Goswami A., Soligo C., Safi K. 2012 Comparative analyses of evolutionary rates reveal different pathways to encephalization in bats, carnivorans, and primates. *Proceedings of the National Academy of Sciences of the United States of America* **109**(44), 18006-18011.
10. Smaers J.B., Vinicius L. 2009 Inferring macro-evolutionary patterns using an adaptive peak model of evolution. *Evolutionary Ecology Research* **11**(7), 991-1015.
11. Felsenstein J. 1988 Phylogenies and quantitative characters. *Annual Review of Ecology and Systematics* **19**, 445-471.
12. Revell L.J. 2012 phytools: an R package for phylogenetic comparative biology (and other things). *Methods in Ecology and Evolution* **3**(2), 217-223.
13. Revell L.J. 2009 Size-correction and principal components for interspecific comparative studies. *Evolution* **63**(12), 3258-3268.